

GwenLaya v4: a pre-registered study of answering only when a check can back the answer

Interim report after the first night run: registered design, deviations, one descriptive single-arm result, and no completed pre-registered test

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Abstract

GwenLaya is a system design in which a small open model (Qwen3.5 family) answers a programming or mathematics question only when an executed check or a calibrated confidence score supports the answer, and otherwise abstains or escalates to a larger tier. We pre-registered fifteen hypotheses (two primary: fewer confident-wrong answers than log-probability selective prediction at matched coverage, H1; lower cost per correct answered item than always using the largest tier with the gate, H3) before any GwenLaya run. This report covers the first, budget-limited night run. **No pre-registered hypothesis test was completed.** The only scored data are 80 items from one arm that is not a registered arm (Qwen3.5-2B, q4_K_M, CPU, no gate, no calibrator). Only the 28 Python items are usable as a descriptive result: 11/28 passed the hidden checks (pass rate 0.393, 95% cluster bootstrap CI 0.214–0.571) while the raw $\exp(\text{mean log-prob})$ confidence averaged 0.896, so it was badly calibrated (ECE 0.519, CI 0.392–0.689); its AUROC was 0.717 with a CI (0.483–0.917) that includes 0.5. These are descriptive numbers on public benchmarks that Qwen3.5 has probably seen in pretraining, not tests. The 26 math items are invalid (a prompt-path defect, and an unregistered last-number answer extraction), and the 26 Rust items come from a harness that has not been validated; numbers pooled over all 80 items are shown only as a labelled sensitivity analysis. The budget hypothesis (H13) is not refuted on an interim basis (known spend \$0.00, 2,706 TPU-seconds unpriced), but the spend ledger enforces it by construction, so this is not evidence about the system. Review also found that the HumanEval+ gate payloads of the night task files contained the hidden tests; no gate ran, so no number is affected, and the files have been rebuilt (D25). We also report a TPU feasibility finding (Qwen2.5 generation works on a v5e chip via vLLM’s TPU build; Qwen3.5 did not load in time; no TPU LoRA path was found for Qwen3.5 dense or Qwen3.8), the GCP spend actually recorded, and every deviation from the registration. We begin by restating the retraction of a v3 claim and the failed v3 gate.

1 Retraction and the v3 gate failure, stated first

Retraction. Revision 3.1.0 of the predecessor report [2] claimed that the low-tier optimizer improved pass rates by at least +8.0 percentage points. That claim rested on a hand-written results file with invented numbers (A0 33%, A3 100%). The file was fabricated, it has been deleted, and it is not evidence (`results/PREREGISTRATION.md`). The claim is retracted. Nothing in this paper depends on it.

The measured v3 gate failed. The pre-registered v3 gate was $A3 - A0 \geq +8.0$ points pass@1 on MBPP-sanitized with `qwen2.5-coder:1.5b`. Both measured runs **FAIL** that gate:

- $n = 60$, CPU: $A3 - A0 = +6.67$ points, 95% CI [1.67, 13.33], 4 gained / 0 lost, exact McNemar $p = 0.125$; verdict FAIL (`results/results_n60_cpu.json`).

- $n = 257$, L4 GPU: A0 64.2 %, A3 71.98 %, $A3 - A0 = +7.78$ points, 95% CI [4.67, 11.28], 20 gained / 0 lost; verdict FAIL against +8.0. This file’s `git_commit` field is empty (`results/results_n257_l4_gpu.json`).

At $n = 60$ the difference is not significant by the exact McNemar test ($p = 0.125$, 4/0 discordant pairs); the percentile-bootstrap interval there is not informative, because when every discordant pair favours A3 a resample can reach $\Delta \leq 0$ only by drawing no discordant item at all, so its lower bound is positive almost by construction. Only the $n = 257$ run (20/0 discordant; the file rounds the exact McNemar p to 0.0, i.e. $p < 0.001$) supports a nonzero effect. In both runs the registered +8.0 threshold was missed, and we report both as FAIL. The same file shows the v3 gate’s weakness that motivates v4: of answers the gate called VERIFIED (coverage 80.93 %), only 88.94 % passed the hidden tests. By arithmetic on those two file values (not a measurement), about 8.95 % of all problems were confident-wrong under v3.

2 What GwenLaya is and what was registered

The full registration is `docs/GWENLAYA_PREREGISTRATION.md` (revision 2, design freeze; sha256 of each revision in `docs/PREREG_FREEZE.txt`). In brief:

- **System (GL).** Generate with a small Qwen3.5 tier; run a *gate-visible* executed check that, by design, never sees the scoring tests (in the night task files this was violated for HumanEval+, whose gate payloads contained the whole hidden suite; no gate ran, and the files were rebuilt, D25, Section 4.3); score the candidate with a calibrated confidence head (a LoRA head on the `convaiinnovations/laya` encoder); answer VERIFIED, LIKELY_CORRECT(p), abstain, or escalate to a larger tier. The label for calibration is the outcome on the *hidden* checks, so VERIFIED is never treated as ground truth.
- **Eval pool E.** HumanEval+ and MBPP+ [11] (Python, 542 items), MultiPL-E Rust [3] (510), MATH-500 [7, 10] (500), and miniF2F Lean 4 (244, outside the pooled analysis).
- **Arms.** B1 smallest tier alone; B2 largest tier alone; B3 log-probability selective prediction [4]; B4 self-consistency; B5 largest tier with the v3-style gate only; B6 Laya zero-shot; LR and a shuffled-label negative control SH.
- **Primary family** (Holm [8], $m = 2$; a primary that does not run enters with $p = 1$, so the other is tested at 0.025): H1, CWR(GL)–CWR(B3) at matched coverage, exact one-sided McNemar [12] plus a cluster-stratified paired bootstrap, MEI 3.0 points; H3, cost per correct answered item, GL vs B5, MEI 25 % reduction, with answered accuracy non-inferior by 2.0 points.
- **Secondary:** calibration (H4 ECE [6], H5 Laya vs LR AUROC, H6 split-conformal selective error [1]); quantization (H8); training (H9); systems (H13 budget, H14 endpoint latency, H15 preemption). **Exploratory:** H2, H7, H10–H12.

Hypothesis wording and MEIs are unchanged. The H3 cost unit (D3), the H5 comparator (D5, if the frozen-embedding probe replaces the LoRA head) and the size of the primary Holm family (D22, made after the L2 generations existed and reverted to the registered $m = 2$ in D26) are deviations. Throughout, VERIFIED keeps its registered meaning (all gate-visible checks passed); the outcome on the hidden checks is written PASS_HIDDEN / FAIL_HIDDEN / UNDECIDED.

3 The night run and its deviations

The night of 2026-10-07/08 was meant to run the primaries cheaply. Two constraints shaped it: the total GCP budget (\$50 hard, \$45 working ceiling for this programme, \$40 planned), and the fact that the project’s only GPU quota unit (GPUS_ALL_REGIONS limit 1) was held all night by a

VM that is not part of this project and that we did not touch. All generation therefore ran on a shared 8-core CPU, and the scope was cut. Every deviation is in `docs/DEVIATIONS.md` and is summarised in Table 1. D1–D20 were written before generation; D21 during L2; D22–D23 after the L2 generations existed and before any registered test; D24–D26 after the external review of the first draft of this report. No registered test has run, so none of these was written with a test result in view.

Table 1: Deviations from the registration for the night run (from `docs/DEVIATIONS.md`). Reason codes: B = budget, G = no GPU slot, T = CPU wall clock, I = implementation or defect.

#	Registered	Night run	Why
D1	E = 1,552 items on L4	E-night: seeded prefix, 80 items (python 28, rust 26, math 26)	G, T
D2	tiers 2B/4B/9B/27B	2B and 4B only (B2/B3/B5 on 4B)	G, T
D3	H3 cost in L4 GPU-seconds	CPU-seconds on this machine	G
D4	separate calibration set C	5-fold cross-fitting on E-night by source cluster	T; C was not label-matched
D5	Laya LoRA r=16 heads	LoRA only if CPU timing fits, else frozen-embedding probe	T
D6	llama.cpp server	llama.cpp b11476 CPU; Ollama not used (no logprobs)	I
D7	Lean 4 (E, C, H11)	dropped tonight	T, B
D8	SFT, DPO, S4b–S5	deferred; H9 deferred, H10–H12 dropped	G, B
D9	B4 self-consistency, second seed	dropped	B, T
D10	GSM8K, AIME25, Live-CodeBench, MBPP anchor, 0.8B	dropped	B, T
D11	H8 on full E	300-item 9B subsample only if GPU; no TOST	G
D12	endpoint, 4–6 cold samples	4 samples, only after G1	G, B
D14–D15	–t 8; 256-token tok/s probe	–t 4; 64 tokens	T (foreign CPU load)
D16	–	fix: multi-letter words in math checker return undecided	I
D17	reference passes its own checks	Rust: typecheck only (no reference solution)	I
D18	decontaminate T, C	all 170 MBPP-sanitized calib items removed; no T/C	I
D19	n from measured throughput	n from measured tok/s and an <i>assumed</i> 196.6 tokens/item	T
D20	prompt text not fixed	gate-visible tests and prompts frozen in the task file	I
D21	–	logprobs and per-call CPU-seconds captured	I
D22	<code>random.Random(0)</code> bootstrap; primary Holm $m = 2$	numpy <code>default_rng(0)</code> , 10,000 resamples; Holm over the primaries that ran ($m = 1$ or 2), a post-data decision-rule change, reverted in D26	I
D23	–	scored the 80 L2 generations; single-arm descriptives added	I
D24	math: boxed answer only; unparseable = UNVERIFIED	scorer fell back to ####, then to the last number in the text; all 26 math items were scored that way; registered boxed-only rescoring now reported too	I (found in re-view)
D25	gate never sees the scoring checks	HumanEval+ gate payloads held the full 5 primary items excluded; no gate ran	I (found in re-view)
D26	Holm $m = 2$	restored ($p = 1$ for a primary that did not run); per-domain descriptives, Python headline	I (post-L2)

Benchmarks reduced. To stay within budget and wall clock we ran no benchmark beyond the three primary domains, on 80 of their 1,541 included items, with two tiers instead of four. Lean 4, GSM8K, AIME25, LiveCodeBench, the MBPP continuity anchor, the 0.8B tier, self-consistency and the second seed were dropped (D7–D10). The registered design for all of them stays in the registration and is not reported here as if it had run.

Power was known to be insufficient before generation. The freeze addendum computed the power of H1’s exact McNemar test at $n = 80$ before any item was generated: 0.052 at the 3-point MEI and 0.261 at a 6-point effect ($\alpha = 0.025$). A non-significant H1 at this n would therefore have been inconclusive, not evidence of no effect.

What actually ran. L0 (environment, throughput), L1 (manifest, freeze addendum, decontamination), L2 (2B generation on all 80 E-night items, 72 minutes wall), and scoring of those 80 generations. L3 (4B generation) was held because the L2 review found a blocking defect (Section 4.3). G1 (GPU) and E1 (endpoint) never ran: the GPU slot never freed. No Laya head, calibrator, threshold or second arm exists.

4 Results

4.1 Per-hypothesis verdicts

Table 2 reproduces docs/RESULTS_VERDICT.md. For every hypothesis that did not run, the probability column is the **red-team prior** written in the review before the run (docs/PROPOSAL_REVIEW.md, copied into docs/DEVIATIONS.md). It is *not* an estimate from data. This includes H13, the only hypothesis whose decision rule could be evaluated: its row shows the prior, because the check holds largely by construction and we give no estimated probability for it.

Table 2: Verdicts after the night run. “Prior” = red-team prior that the hypothesis would be supported, written before any data; not updated by evidence.

H	Role	Verdict	$P(\text{true})$	Basis
H1	primary	not run	0.45 (prior)	no GL or B3 arm; power at MEI with $n = 80$ was 0.05
H3	primary	not run	0.50 (prior)	no GL or B5 arm; cost unit would be CPU-s (D3)
H4	secondary	not run	0.15 (prior)	no cross-fitted scores; no Laya head
H5	secondary	not run	0.25 (prior)	same
H6	secondary	not run	0.40 (prior)	same
H8	secondary	not run	0.30 (prior)	needs GPU (D11)
H9	secondary	deferred	0.25 (prior)	D8
H13	systems	not refuted (interim)	0.75 (prior)	known spend \$0.00; 2,706 TPU-s unpriced; holds by construction, Sec. 6
H14	systems	not run	0.50 (prior)	needs E1, which needs the GPU (D12)
H15	systems	not run	0.70 (prior)	offline test is not the registered SIGKILL check
H2	exploratory	not run	0.08 (prior)	no GL or B5 arm
H7	exploratory	not run	0.10 (prior)	gate field is null in every scored row
H10	exploratory	dropped	0.05 (prior)	D8
H11	exploratory	dropped	0.12 (prior)	D7, D8
H12	exploratory	dropped	0.12 (prior)	D8

H13 (budget). Decision rule: ledger total $\leq \$40$ planned and $\leq \$50$ hard, and no stage over $1.25\times$ its cap twice. The machine verdict (`numbers_v4.json`) is “not refuted on the known lines (lower bound; unpriced lines exist)”: the ledger has 3 lines with a known USD total of 0.0, and 2 lines (2,706 seconds of TPU time) carry no `usd_estimate` and no `resource` field, so they break the ledger schema and are unpriced. We do not price them here (TBD until the billing export is read), and no stage has spent anything against its cap. Two caveats limit what this means. First, the ledger refuses any stage that would exceed its cap, so H13 holds largely by construction; the red-team review called its support “close to tautological” (prior 0.75), and it is not evidence about the GwenLaya system. Second, H13 is a whole-study claim, the planned GCS ledger `runs/LEDGER.jsonl` does not exist, and later nights will add spend, so the verdict is interim. An earlier draft of this report gave $P(\text{true}) = 0.97$; that was author judgement with no statistical basis, and it has been removed.

H15 (preemption). `tests/test_run_study.py::test_resume_after_kill_mid_run` passes, but it simulates a death by appending a torn line; it sends no SIGKILL and measures no lost wall time. It is weak support that resume is item-granular, not the registered verification, so the probability stays at the prior.

4.2 Descriptive single-arm result (not a pre-registered test)

The scored arm `base` is Qwen3.5-2B q4_K_M on llama.cpp [5], greedy, thinking off, `max_new_tokens` 1024, with no gate, no calibrator and no escalation. It is not one of the registered arms. Confidence is the raw $\exp(\text{mean token log-prob})$, uncalibrated and not cross-fitted. Intervals are 95% cluster-stratified bootstrap percentile intervals (10,000 resamples, seed 0). Every number below is read from `papers/numbers_v4.json` (`single_arm.by_domain.*`, `single_arm.boxed_only.*`, `arm.base.*`), written by `scripts/analyze_study.py`; its tables are reproduced unedited in Appendix A.

Outcome labels. The per-row outcome reported here is the result of the *hidden* checks: PASS_HIDDEN, FAIL_HIDDEN or UNDECIDED (`rows.jsonl` field `score` = VERIFIED, FAILED or UNVERIFIED). It is *not* the registered VERIFIED, which is the gate’s verdict on the gate-visible checks. The gate verdict was not recorded in any of the 80 rows (`gate` is null in all of them), so this run contains no VERIFIED in the registered sense.

Why Python is the headline. Pooling the three domains would mix one usable domain with two that are not (Section 4.3): the 26 math generations are invalid, and the 26 Rust items come from an unvalidated harness on which no item passed, so their AUROC is undefined. A pooled AUROC or ECE would then describe domain composition and defects, not the confidence signal. We therefore report Python ($n = 28$) as the headline and show every other number only as a labelled row of Table 3.

Python. 11 of 28 items passed the hidden checks and 17 failed (0 undecided), a pass rate of 0.393 (CI 0.214–0.571). The mean raw confidence was 0.896 (CI 0.875–0.917), so the score is badly calibrated (ECE 0.519, CI 0.392–0.689; Brier 0.476, CI 0.334–0.617). The AUROC point estimate was 0.717 (AURC 0.438, CI 0.221–0.712), but its interval (0.483–0.917) includes 0.5, so with $n = 28$ we cannot say that it ranks correct answers above wrong ones. The confident-wrong rate, counting every answered item that did not pass, was 0.607 (CI 0.429–0.786). Cost per passing item was 330 server CPU-seconds (CI 174–726).

Rust and math (not interpretable). Rust: 0 PASS_HIDDEN, 13 FAIL_HIDDEN, 13 UNDECIDED. Math: 3 PASS_HIDDEN, 18 FAIL_HIDDEN, 5 UNDECIDED as scored; all 26 math answers were extracted by the unregistered last-number fallback (0 contained `\boxed`), including

Table 3: Raw confidence of arm **base**, descriptive (from `papers/numbers_v4.json`). Pass = PASS_HIDDEN. 95% cluster bootstrap CI in brackets. Only the Python row is a headline; the other rows are sensitivity analyses and are not valid estimates. Cost = server CPU-seconds per passing item (D3, D21). TBD = undefined (no passing item).

Subset	n	Pass rate	Mean conf.	ECE	AUROC	Cost / pass
Python (headline)	28	0.393 [0.214, 0.571]	0.896 [0.875, 0.917]	0.519 [0.392, 0.689]	0.717 [0.483, 0.917]	330 [174, 726]
<i>Sensitivity only, not valid estimates</i>						
Rust (unvalidated)	26	0.000	0.894 [0.882, 0.906]	0.894 [0.882, 0.906]	TBD	TBD
Math (invalid), as scored ^a	26	0.115 [0.000, 0.231]	0.832 [0.807, 0.857]	0.734 [0.633, 0.833]	0.348 [0.133, 0.583]	3304 [1298, 11975]
Math (invalid), boxed-only ^b	26	0.000	0.832 [0.807, 0.857]	0.832 [0.807, 0.857]	TBD	TBD
Pooled, as scored	80	0.175 [0.100, 0.250]	0.875 [0.863, 0.886]	0.700 [0.626, 0.771]	0.618 [0.422, 0.811]	1178 [732, 2114]
Pooled, boxed-only math	80	0.138 [0.075, 0.200]	0.875 [0.863, 0.886]	0.737 [0.676, 0.798]	0.758 [0.568, 0.922]	1500 [931, 2840]

^a Unregistered extraction: last `\boxed`, else `####`, else the last number in the text (D24). ^b Registered rule: no `\boxed` = UNVERIFIED.

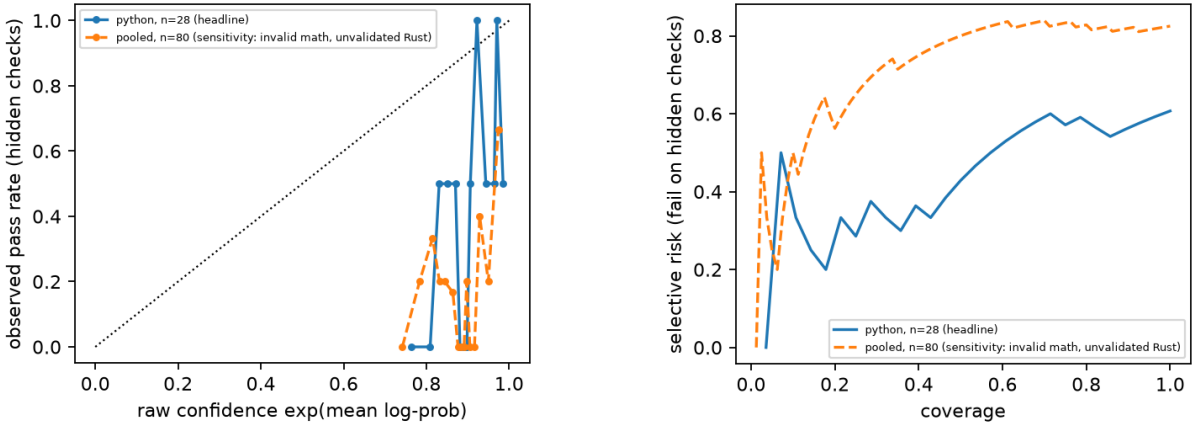


Figure 1: Arm **base**, raw confidence. Solid: Python, $n = 28$ (headline). Dashed: all 80 items, a sensitivity curve only, because it contains the invalid math items and the unvalidated Rust items (Section 4.3). Left: reliability diagram (equal-mass bins). Right: risk-coverage curve; risk = share failing the hidden checks. Descriptive only; not cross-fitted.

all 3 passes, and under the registered boxed-only rule math passes 0/26. Pooled over 80 items the pass rate is 0.175 as scored and 0.138 with boxed-only math. The pooled AUROC changes from 0.618 to 0.758 when only the scoring rule of the invalid math subset changes, which is why we do not interpret it.

Contamination. HumanEval+, MBPP+, MultiPL-E and MATH-500 are public and were very probably seen in Qwen3.5 pretraining (registration Section 10, “Residual risk, stated”). The absolute pass rates above are therefore not clean capability estimates; the registration claims only paired, within-model deltas, and none was measured. Memorised items would also tend to have high log-probability and pass, which can inflate AUROC differently by score type (review finding F12).

What we read from this, and no more: on 28 Python items the raw log-probability of a 2B model at q4 sits near 0.9 while about 0.4 of the answers pass, so it is not usable as a probability without calibration. This is consistent with the motivation for a calibrated head, but it does not test H1, H4 or H5.

4.3 Defects found, and why some numbers are invalid

1. **Math prompt path (blocking).** All 26 math completions were produced with the Python code system prompt and a `python` prefill, because the prompt builder falls back to the `python` tag for any domain it does not list. 0 of 26 contain `\boxed`. The frozen user prompt asked for a boxed answer, but the system prompt and prefill overrode it. The math pass rate says nothing about the model’s math ability. The fix (math system prompt, no code prefill) and a regeneration of the 26 items must be logged as a new deviation before any rerun; neither has happened yet.
2. **Unregistered math answer extraction (D24).** The registration scores the boxed answer and treats unparseable math output as UNVERIFIED (fail-closed). The scorer (`math_check.extract_final_answer`) instead tries the last `\boxed`, then a `####` line, then the last number anywhere in the text. With no boxed output, all 26 math items were scored by the last number; the 3 passes came from text ending in Python comments, a `print` call or a `return` statement. We now report both versions (Table 3) and froze the registered boxed-only rule before the math regeneration (`docs/FREEZE_ADDENDUM_2026-10-08_D25.md`).
3. **Rust harness unvalidated.** MultiPL-E has no reference solutions, so the Rust harness was checked only by typechecking test bodies (D17). 0/26 PASS_HIDDEN with 13/26 UNDECIDED is as consistent with a harness or toolchain problem as with model weakness. Until one known-good Rust program passes end to end, the Rust rate is unverified.
4. **Gate not recorded.** The `gate` field is null in every scored row, so no registered VERIFIED exists in this run and VERIFIED precision against hidden checks (H7) cannot be computed.
5. **Hidden tests in the HumanEval+ gate payload (D25).** The task builder restricted the HumanEval+ gate by rewriting only the loop header (`zip(inputs[:1], results[:1])`), so the complete hidden `inputs` and `results` lists stayed in every HumanEval+ gate payload (11/11 E-night rows, 161/161 primary rows; four primary items also carried a reference implementation). A candidate executed in the same process as `check()` could have read them, and any component that reads the gate payload would have seen the hidden answers. MBPP+, Rust and math gate payloads were correctly restricted. The statement that the gate “never sees the scoring checks” was therefore false for HumanEval+ in the night files. **No reported number changes:** the L2 run executed no gate. The builder now rewrites the literals themselves with an AST transform, fails closed, and is covered by a test asserting that no hidden element appears in a gate payload. The task files were rebuilt with prompts and hidden checks unchanged (E-night 80 items; primary E 1,536 items after 5 fail-closed exclusions; hashes in Section 7).

What “passed” means in this paper. An item is PASS_HIDDEN only if it passed the named hidden checks: for Python, the full EvalPlus test suite run in a bubblewrap sandbox; for Rust, compilation and all `assert_eq!` lines of the MultiPL-E test with `rustc`; for math, a sympy/numeric equivalence check between the gold answer and the answer extracted as described in item 2 above (boxed, else `####`, else last number). It means nothing more. It does not mean the solution is correct on other inputs, and for Rust the harness itself is not yet validated.

5 TPU feasibility finding

Because the GPU slot was unavailable, we tested whether a Cloud TPU v5e could replace it. Two smoke runs were made on a single-chip v5litepod-1 (logs under `gs://socrateai-datalake-gen-lang-client-0625573011/gwenlaya_v4/tpu_smoke/gwenlaya-tpu-smoke-215529/`). The first, on a spot VM, was preempted after about 4 minutes with no results. The second, on demand:

- **Qwen2.5 works.** Qwen/Qwen2.5-Coder-1.5B-Instruct loaded and generated with vLLM [9] 0.31.0's TPU backend (load 208.9 s; 442 completion tokens in 0.61 s, 724.6 tok/s batched throughput; `res_qwen25.json`). The GWAYA gate on that VM returned UNVERIFIED for every item because bubblewrap was not available there, i.e. it failed closed as designed; no correctness number came out of this run.
- **Qwen3.5 is unproven on TPU.** Qwen/Qwen3.5-4B resolved to Qwen3_5ForConditionalGeneration but did not finish loading/compiling within the 1,500 s step limit (exit 124). We did not show that it works or that it cannot work.
- **No TPU LoRA path was found for Qwen3.5 dense or Qwen3.8.** Serving Qwen2.5 with `enable_lora` failed at engine initialisation. The JAX training stacks we inspected list Qwen2/Qwen3 (Tunix 0.1.7) and Qwen2.5/Qwen3 plus Qwen3.5 mixture-of-experts scripts (MaxText), but no dense Qwen3.5 or Qwen3.8 path. GwenLaya's training steps (Laya head, SFT) therefore stay on the GPU/CPU plan.

6 Actual GCP spend

Read from `/mnt/data/home/xavkal/gwaya-data/spend_ledger.jsonl` (3 lines):

Time	Item	Seconds	USD in ledger
2026-10-07 21:54	TPU smoke, attempt 1 (v5litepod-1 spot, us-west4-a; preempted)	240	not recorded
2026-10-07 22:38	TPU smoke (v5litepod-1 on demand, us-central1-a)	2,466	not recorded
2026-10-08 00:35	L1 upload to data lake (6 objects)	13	0.0

The recorded USD total is \$0.00 with 2,706 seconds of TPU time unpriced. The two TPU lines have neither a `usd_estimate` nor a `resource` field, so they do not meet the ledger schema (`ts`, `item`, `usd_estimate`, `seconds`, `resource`). The planning documents carry a figure of about \$0.84 for that TPU time; it comes from the task description, not from the ledger, a price list or a bill, and no verdict in this paper uses it. The night's GPU (\$2.00) and endpoint (\$2.00) caps were not used. Local CPU work cost nothing on GCP. The billed amount is TBD until the billing export is read.

7 Reproducibility

Code. Repository branch `night/2026-10-07` (from `main`, released as v3.2.0 at 825d1ba); the first draft of this paper is commit 9b78e18; `numbers_v4.json` was regenerated for this revision (all numbers of the first draft unchanged, per-domain and boxed-only numbers added) and is committed with it. llama.cpp release b11476 (CPU, asset sha256 2cda5ff9...4b5e); Qwen3.5-2B-Q4_K_M.gguf from unsloth/Qwen3.5-2B-GGUF revision f6d5376b..., sha256 aaf42c8b...9223; 4B: revision e87f1764..., sha256 00fe7986...11a4 (all in `L0_env.json`).

Frozen inputs (sha256).

File	sha256
experiments/night/manifest_eval.json	bd1684dadcf38328fb6fab7619e5ea2f4fbc5ccc0e31022e059360b2d8c615e0
tasks_E_night.jsonl (80 items)	e842edd13951b4082c4495f7bcc845b032989f4a457e0c5d6e9c44f2b72cd679
tasks_E_primary.jsonl (1,541 items)	971f44c61bad5044f8487383f8a3ea27bc2c292c565a6ff8ef1254fce1cc7097
docs/FREEZE_ADDENDUM_NIGHT_2026-10-07.md	ac2970983ef45eb027de65194d42cadd41483dfd35cd2a866efc64247b02376c
decontam_report.json	d128e06c5949c86c29537f71a1d547ef953be6aa22f7bc814a280ec4d48f1907
night/cache/L2score/rows.jsonl	1e35fac792d29d8f82ce05f5cbad0d615bb4db2a6a707d157edff75da0fa69c6
night/cache/L2score/gens.jsonl	03f6e2b6af9cebd65d8488b9a35a1f7d3bc70b566397379edc3d95f5291963db
papers/numbers_v4.json (this revision)	e5488f6c66419aa99971eef25c10f1f076b0595d1b9c02db84e007591d32b07c
tasks_E_night.d25.jsonl (80, D25)	504909fc74dc2af8fb3f2491a475df07de8e95e6fdf0f4f69bc22352e95e0952
tasks_E_primary.d25.jsonl (1,536, D25)	2143110c78a3a9f57599dce77021cf4571fc57cf2584017f16ebdd08d5acba66

HF dataset revisions are in the freeze addendum (MATH-500 6e4ed1a2, humanevalplus d32357cf, mbppplus b2d74c91, MultiPL-E 28441b60). The rows.jsonl and gens.jsonl hash chains verify (numbers_v4.json meta). The L2 run used the original task files; the .d25 files (gate payloads rebuilt, prompts and hidden checks unchanged) are for all later runs and are described in docs/FREEZE_ADDENDUM_2026-10-08_D25.md.

Data lake. `gs://socrateai-datalake-gen-lang-client-0625573011/gwenlaya_v4/night/2026-10-07/` holds the manifest, both task files, the sizing file, the addendum and the decontamination report. The TPU logs are under `.../gwenlaya_v4/tpu_smoke/`. The scored L2 rows and generations are *not yet* in the lake (they are on the local data disk under `night/cache/L2score/`); the registration requires results tables to come from `gwenlaya_v4/results/`, so uploading them is an open item and this is recorded as a limitation.

Commands. With the environment of `night/run_L2.sh` (GWAYA_DATA_ROOT, HF_HOME, PYTHONPATH):

```
python scripts/run_study.py --plan experiments/night/plan_night.json --stage night_L2 \
  --backend openai --backend-url http://127.0.0.1:8091/v1 \
  --tasks $NIGHT/results/tasks_E_night.jsonl --models qwen3.5-2b-q4_K_M \
  --quants q4_K_M --arms base --mode generate --no-vram --cpu-pid <llama-server pid> \
  --out $NIGHT/cache/L2
python scripts/run_study.py ... --mode score --out $NIGHT/cache/L2score
python scripts/analyze_study.py --results $NIGHT/cache/L2score --n-boot 10000 --seed 0
# writes numbers_v4.json, tables_v4.tex, figures_v4/; --results must name L2score only,
# because night/cache/L2/gens.jsonl is a byte-identical copy and would be counted twice
python scripts/data/regen_gate_payloads_d25.py --results $NIGHT/results # D25 task files
```

The llama-server ran on 127.0.0.1:8091 with `-t 4 -parallel 2` (D14).

8 Limitations

- No pre-registered test ran. Nothing here supports or refutes H1–H12, H14 or H15.
- The only scored arm is unregistered, single-seed, $n = 80$, on CPU, with no gate recorded.

- The math generations are invalid (prompt-path defect) and were scored with an unregistered last-number extraction (D24); the Rust harness is unvalidated. Only the 28 Python items are a usable descriptive result, and with $n = 28$ their AUROC interval includes 0.5.
- **Contamination.** All absolute scores are on public sets (HumanEval+, MBPP+, MultiPL-E, MATH-500) that Qwen3.5 has probably seen in pretraining; they are not clean capability estimates, and only paired within-model deltas are claimed by the registration. Decontamination (D18) covered only training/calibration data against E and cannot detect pretraining exposure. The contamination-stratified analysis asked for by review finding F12 (for example MBPP+ versus HumanEval+ items, or a post-cutoff anchor such as LiveCodeBench, which D10 dropped) was not done: TBD.
- The HumanEval+ gate payloads of the night task files contained the hidden tests (D25). No gate ran in L2, so no result is affected, but any earlier claim that the gate never saw the scoring checks was false for those files.
- Several deviations (D22–D26) were written after the L2 generations existed; none was written after a registered test, because none ran.
- Even with all arms present, $n = 80$ gives H1 power of about 0.05 at its MEI; the night design could at best produce an inconclusive H1.
- Cost is server CPU-seconds on a machine shared with foreign jobs and includes idle threads; it is not comparable to the registered L4 GPU-seconds.
- Two ledger lines are unpriced and lack the **resource** field; the USD spend above is a lower bound, and H13 holds largely by construction of the ledger.
- Scored rows are not yet mirrored to the data lake as the registration requires.
- Lean 4, the training arms, quantization and the endpoint were not run. The TPU finding is from one chip, one vLLM version and a fixed time limit.

9 Next steps

Log the math prompt fix as a deviation, regenerate the 26 math items and score them boxed-only (D24), price the two TPU ledger lines from the billing export, validate the Rust harness on one passing program, generate the 4B arm and the GL/B3/B5 arms on the same 80 items with the .d25 task files, build the cross-fitted scores file, upload results to the data lake, and rerun `analyze_study.py`. Only then can H1, H3, H2, H7 and H4–H6 move off “not run”. A null or negative result will be reported in the same form as this one.

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A Machine-generated tables

The tables on the next page are `papers/tables_v4.tex` as written by `scripts/analyze_study.py` from `papers/numbers_v4.json` (content unedited, font reduced). “CPU-s total” is server CPU-seconds, which include idle-spinning worker threads on a machine shared with three foreign CPU-bound jobs (D14, D21). In these tables “acc.” and “pass rate” are the PASS_HIDDEN rate (rows.jsonl `score` = VERIFIED, i.e. passed the hidden checks), not the registered gate VERIFIED. CWR counts every UNDECIDED answer as wrong, so it is an upper bound. The arm table holds the pooled 80 items and is a sensitivity analysis only (Section 4). The hypothesis-status table is the script’s machine status; Table 2 is the reviewed verdict.

Table 4: Cached generations on the E-night set (greedy, one seed). Tokens are mean completion tokens.

model	domain	n	mean tok.	CPU-s total	finish reasons
qwen3.5-2b-q4_K_M	math	26	371.0	9911	length:4, stop:22
qwen3.5-2b-q4_K_M	python	28	118.8	3625	stop:27, length:1
qwen3.5-2b-q4_K_M	rust	26	98.5	2963	stop:26

Table 5: Single unregistered arm **base**, per domain (descriptive, not a test). Pass = passed the hidden checks (PASS_HIDDEN), not the registered gate VERIFIED. Confidence = raw exp(mean log-prob). 95% cluster bootstrap CI. Cost unit: cpu_seconds. TBD = undefined (e.g. no passing item).

domain	status	n	pass rate	mean conf.	ECE	AUROC	cost / pass
math	INVALID prompt path; last-number fallback	26	0.115 [0.000, 0.231]	0.832 [0.807, 0.857]	0.734 [0.633, 0.833]	0.348 [0.133, 0.583]	3304 [1298, 11975]
python	headline	28	0.393 [0.214, 0.571]	0.896 [0.875, 0.917]	0.519 [0.392, 0.689]	0.717 [0.483, 0.917]	330 [174, 726]
rust	unvalidated harness	26	0.000 [0.000, 0.000]	0.894 [0.882, 0.906]	0.894 [0.882, 0.906]	TBD [TBD, TBD]	TBD [TBD, TBD]
math (boxed-only)	INVALID prompt path; registered boxed-only	26	0.000 [0.000, 0.000]	0.832 [0.807, 0.857]	0.832 [0.807, 0.857]	TBD [TBD, TBD]	TBD [TBD, TBD]
pooled (boxed-only math)	sensitivity	80	0.138 [0.075, 0.200]	0.875 [0.863, 0.886]	0.737 [0.676, 0.798]	0.758 [0.568, 0.922]	1500 [931, 2840]
pooled	sensitivity (mixed)	80	0.175 [0.100, 0.250]	0.875 [0.863, 0.886]	0.700 [0.626, 0.771]	0.618 [0.422, 0.811]	1178 [732, 2114]

Table 6: Arms on the paired E-night items (percent; 95% cluster bootstrap CI). Cost unit: cpu_seconds. TBD = not run or not available.

arm	acc.	coverage	CWR	ans. acc.	cost / correct
base	17.5 [10.0, 25.0]	100.0 [100.0, 100.0]	82.5 [75.0, 90.0]	17.5 [10.0, 25.0]	1178.4 [731.8, 2113.9]

Table 7: Hypothesis status for the night run. Verdicts follow the pre-registered rules; primary Holm with $m = 2$.

H	role	result	detail
H1	primary	TBD	scored rows exist but only for unregistered single arm(s) ['base']: H1/H2/H3/H7 need paired registered arms (GL, B1, B2, B3, B5)
H2	exploratory	TBD	scored rows exist but only for unregistered single arm(s) ['base']: H1/H2/H3/H7 need paired registered arms (GL, B1, B2, B3, B5)
H3	primary	TBD	scored rows exist but only for unregistered single arm(s) ['base']: H1/H2/H3/H7 need paired registered arms (GL, B1, B2, B3, B5)
H4	secondary	TBD	no cross-fitted scores file (no scores file)
H5	secondary	TBD	no cross-fitted scores file (no scores file)
H6	secondary	TBD	no cross-fitted scores file (no scores file)
H7	exploratory	TBD	scored rows exist but only for unregistered single arm(s) ['base']: H1/H2/H3/H7 need paired registered arms (GL, B1, B2, B3, B5)
H8	exploratory_only_if_G1	TBD	needs G1 (no GPU slot)
H9	deferred	deferred	deferred (D8)
H10	dropped	dropped	dropped for this run (docs/DEVIATIONS.md D8)
H11	dropped	dropped	dropped for this run (docs/DEVIATIONS.md D8)
H12	dropped	dropped	dropped for this run (docs/DEVIATIONS.md D8)
H13	run	not refuted on the known lines (lower bound; unpriced lines exist)	
H14	only_if_E1	TBD	needs E1 (D12)
H15	offline_test	TBD	offline kill/resume test, see tests/test_run_study.py